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Modelling the complexity of the foot and ankle during human locomotion: the development and validation of a multi-segment foot model using biplanar videoradiography

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ABSTRACT

We developed and validated a multi-segment foot and ankle model for human walking and running. The model has 6-segments, and 7 degrees of freedom; motion between foot segments were constrained with a single oblique axis to enable triplanar motion [Joint Constrained (JC) model]. The accuracy of the JC model and that of a conventional model using a 6 degrees of freedom approach were assessed by comparison to segment motion determined with biplanar videoradiography. Compared to the 6-DoF model, our JC model demonstrated significantly smaller RMS differences [JC: 2.19° (1.43–2.73); 6-DoF: 3.25° (1.37–5.89)] across walking and running. The JC model is thus capable of more accurate musculoskeletal analyses and is also well suited for predictive simulations.

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KEYWORDS
Multi-segment foot model; biplanar videoradiography; kinematics; modelling; OpenSim

Introduction

In the human foot, 26 bones move with respect to each other around 33 articulations that are spanned by approximately 100 muscles and ligaments. Accurately measuring the motions of these bones, and the impact of these motions on soft-tissue deformation is critical for our understanding of foot function, human locomotion, as well as cause-and-effect relationships associated with lower limb pathologies. The most common approach uses optical motion capture of markers fixed to the skin surface to determine the motion of multiple segments relative to each other – typically termed multi-segment foot modelling. Such models simplify the complex anatomy of the foot to understand skeletal foot function, however these models are only used to imply bone kinematics and are less useful for understanding or simulating musculoskeletal dynamics and neuromuscular control of the foot. Therefore, a model that can capture the musculoskeletal complexity of the foot and can accurately measure joint function may enhance our ability to predict outcomes of orthopaedic surgeries, optimise assistive devices and assess mechanisms of musculoskeletal injuries.

There are numerous multi-segment foot models in the literature, with varying degrees of complexity (ranging from 2 to 26 segment models) (MacWilliams et al. 2003; Leardini et al. 2007; Nester et al. 2007; Oosterwaal et al. 2016), however they have a number of clinical and technical challenges that need to be overcome. First, to evaluate functional abnormalities of the foot, motion of the talus bone is essential, both in relation to the shank and calcaneus. Attaching markers on the talus during gait is not feasible due to its anatomical location deep in the ankle mortise. A global optimisation (inverse kinematics) approach with joint constraints (JC) may help overcome this issue, as generalised coordinates of each segment are computed by placing the model in a pose which aligns with experimental markers at each time step (Lu and O'Connor 1999). However, no study to date has assessed how accurately an inverse kinematics solution is able to track the talus and compute joint kinematics. Furthermore, from a clinical and evolutionary

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perspective, the medial longitudinal arch is perhaps one of the most important features of the foot. Quantification of medial longitudinal arch function requires accurate computation of midfoot and forefoot kinematics, particularly in the sagittal plane (Caravaggi et al. 2019). However, most multi-segment foot models may oversimplify these segments and 6-DoF model calculations involve violation of rigid body assumptions such that bone segments may intersect (Leardini et al. 2005). A 6-DoF model also presents challenges when performing more complex biomechanical analysis such as forward simulations, as solving the redundancy problem is much tougher in a multi-segment foot model with 6-DoF at each joint.

In contrast, an articulated model that links rigid body segments *via* joints that constrain motion of one segment relative to another about a centre pivot (centre of rotation) and an axis angled relative to all three cardinal body planes may better model joint motion in the foot. Morphologically relevant JCs with global optimisation (inverse kinematics) have been shown to compensate for skin movement artefact and reduce errors particularly in the frontal and transverse planes (Lu and O'Connor 1999). While, such constraints also potentially conceal some physiologically relevant joint motion (e.g. joint translations), Malaquias et al. (2017) have previously shown that a foot and ankle model with reduced DoF produced more realistic results compared to a model with more DoF, when bone pin data is used as gold standard (Lundgren et al. 2008). Joint constrained models are also better suited for simulations, allowing us to investigate questions that are not possible using experimental techniques such as the force generation capacity of muscles during gait and the design of prostheses and exoskeletons (Farris et al. 2014). However, understanding the accuracy of these JCs and their influence on kinematic gait measures is important and remains unclear (Kainz et al. 2016).

Regardless of the model used, one difficulty in modelling motion of the foot is confirming the validity of the assumptions made in the model. The performance of some multi-segment foot models have been compared to bone pin data, and have demonstrated substantial mean differences in joint kinematics ranging from 3° to 8° (Nester et al. 2007). These differences are particularly alarming as some joints in the foot only have a total rotational range of 15°–20°. However, it is unclear whether the bone pin approach represents a true gold standard measure as it is invasive and soft-tissue movement can often cause pin bending, and thus errors in kinematics. Alternatively, biplanar videoradiography (BVR) is able to provide unique insight into the intrinsic function of the foot with high resolution and accuracy. Not bound to the limitations of bone pins and optical motion capture, BVR combines high-speed X-ray image sequences with 3D bone volumes obtained from computed tomography (CT) scans to track 3D bone movement of individual bones. With its excellent measurement accuracy, within 1 mm in translation and 1° in orientation in the foot (Wang et al. 2015) and high resolution, BVR has great potential to help develop and assess the accuracy of multi-segment foot models.

Therefore, the aim of this study was to develop and validate an articulated JC multi-segment foot and ankle model and to compare the accuracy to a more conventional 6-DoF modelling approach (6-DoF model). Development of the model required definitions of JCs established from the literature to be implemented into the model such that an inverse kinematics approach could be used to model the motion. To determine each models' accuracy (JC and 6-DoF), we computed the joint kinematics from each model during both walking and running and compared this to motion measured simultaneously using BVR. Walking and running were examined to determine the consistency in model outputs across different gaits. To test the robustness of the JC model we also assessed its sensitivity to deviations in joint parameters, particularly position of the centre of rotation and the orientation of the joint axes. We hypothesised: (i) a JC model would have comparable if not smaller deviations from BVR compared to a 6-DoF model; (ii) differences at the ankle, subtalar, midtarsal and tarsometatarsal joints would be smallest in the sagittal plane and (iii) differences would be most sensitive to deviations in the position of the joint centre of rotation compared to changes in joint axes orientation. The model was developed in OpenSim (3.3) and is available from SimTK.org (https://simtk.org/projects/footankle_model) for analysis of optical motion capture for human walking and running.

Model and methods

A constrained multi-segment model of the foot (JC)

Rigid body segments

We used bone geometry available in OpenSim software (Delp et al. 2007) to create six rigid body segments: tibia, talus, calcaneus, midfoot, forefoot and digits. The marker set used to scale and track the segments was based on the Rizzoli foot model (Leardini et al. 2007) as it has been shown to have a high inter- and intra-tester repeatability in healthy adults

Table 1. Description of the body segment marker locations.

Segment	Name	Description
Knee	MKN	Marker placed on medial epicondyle of the knee
	LKN	Marker placed on lateral epicondyle of the knee
Shank	S 1-3	Cluster placed on anterior aspect of the leg
Ankle	MM	Marker placed on medial malleolus
	LM	Marker placed on lateral malleolus
Rear-foot	PC	Marker placed on superior aspect of posterior calcaneus
	DC	Marker placed on inferior aspect of posterior calcaneus
	MC	Marker placed on sustentaculum tali
	LC	Marker placed on peroneal tubercle
Midfoot	NAV	Marker placed on navicular tuberosity
	MB1	Marker placed on base of the first metatarsal
	MB2	Marker placed on base of the second metatarsal
	MB5	Marker placed on base of the fifth metatarsal
Forefoot	MH1	Marker placed on head of the first metatarsal
	MH2	Marker placed on head of the second metatarsal
	MH5	Marker placed on head of the fifth metatarsal
Digits	D1	Marker placed on the proximal phalanx of the hallux
	D2	Marker placed on the proximal phalanx of second digit
	D5	Marker placed on the proximal phalanx of fifth digit

(Caravaggi et al. 2011). The location of the markers attached to each segment are detailed in Table 1. The midfoot represents motion of the navicular, cuboid, medial cuneiform, intermediate cuneiform and lateral cuneiform and the forefoot tracks the movement of the five metatarsal bones.

We aligned the coordinate system of each rigid body such that when the model is in the anatomical position, the *x*-direction pointed anteriorly, the *y*-direction pointed superiorly and the *z*-direction to the right. The talus coordinate system was located at the midpoint between the medial and lateral malleoli markers, at vertical centre of the bone. The calcaneus coordinate system was located at the midpoint between the medial and lateral calcaneal markers, at vertical centre of the bone. The midfoot coordinate system was located at the midpoint between the navicular and fifth metatarsal base at the proximal aspect of the segment. The forefoot coordinate system was located at the second metatarsal base while the coordinate systems for the digits was located at the second proximal phalanx.

Whilst not required for the purposes of kinematic modelling, the inertial properties of each segment were assigned to enable future use in inverse dynamic calculations. The mass, centre of mass and moments of inertia of each foot segment were calculated by representing their volume as a cone (Leardini et al. 2007), with the dimensions derived from the landmarks on each segment, as previously described by Leardini et al. (2007).

Joints

The five joints connecting the segments were the ankle (talus–tibia), subtalar (calcaneus–talus), midtarsal (midfoot–calcaneus), tarsometatarsal (forefoot–midfoot) and the metatarsophalangeal joints (digits–forefoot). The JC model constrained motion between rigid body segments by linking them *via* rotational joints. This allowed segments to only rotate about the defined axis at a centre of rotation and thus with 1 DoF. The axis of each joint in the foot was orientated relative to all three cardinal planes, allowing for simultaneous 3D rotations, triplane motion. The orientation of each joint axis was defined by an inclination and deviation angle as reported by previous cadaveric studies (Hicks 1953; Nester et al. 2001; Lewis et al. 2007), although slightly modified during the iterative model building process. The inclination angle was defined as the angle between the sagittal projection of the axis and the horizontal plane and the deviation angle was along the horizontal projection, and measured with respect to the long axis of the foot (straight line between calcaneus and second metatarsal) (van den Bogert et al. 1994). The joint axis at the subtalar had a 38° inclination and −21° deviation to the midline of the body, running from posterior, inferior and lateral to anterior, superior and medial through the rear-foot (Lewis et al. 2007). The midtarsal joint axis was orientated in an upward, medial direction with 35° of inclination and −24° of deviation to the midline of the body (Nester et al. 2001). The tarsometatarsal joint axis had an inclination of 36° and deviation of −60° from the midline of the body, which runs in a medial, upward direction from the fifth to the second metatarsal (Hicks 1953). These joint axes are shown in Figure 1.

The 6-DoF model was similar to the Rizzoli foot model, consisting of five segments, tibia, talus/calcaneus, midfoot, forefoot and digits (Leardini et al. 2007). The motion between the talus and calcaneus was fixed. The interaction of two adjacent segments was modelled such that they could rotate and translate about all three cardinal planes of motion

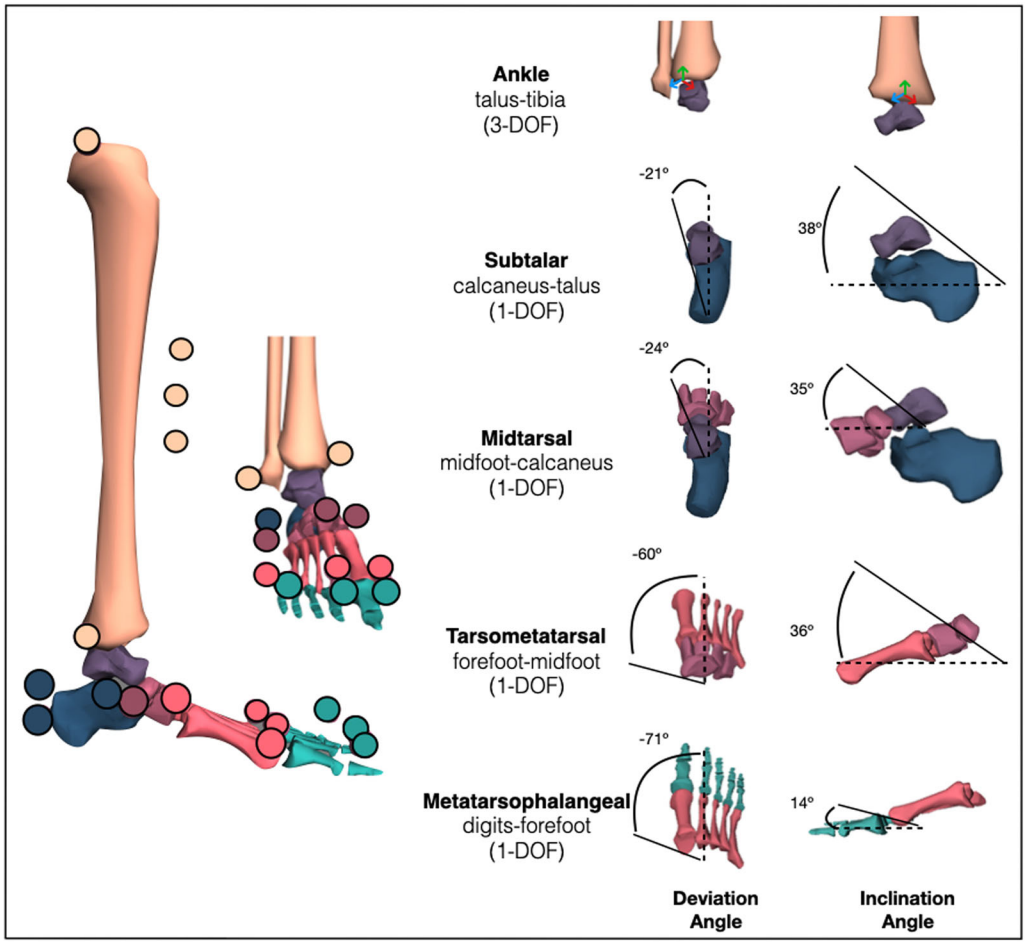


Figure 1. The joint constrained (JC) multi-segment foot model consisted of six segments (tibia, talus, calcaneus, midfoot, forefoot and digits) with a total of 7 degrees of freedom (DoF); all joints except for the ankle were model as a revolute joint with a single oblique axis; inclination and deviation angles for each joint are illustrated. The inclination angle was defined as the angle between the sagittal projection of the axis and the horizontal plane, with the deviation angle being along the horizontal projection and measured with respect to the long axis of the foot (straight line between calcaneus and second metatarsal). The Rizzoli marker set (Leardini et al. 2007) was used to scale the segments of the model and compute joint kinematics using inverse kinematics analyses.

(3× rotations + 3× translations). The primarily difference between our 6-DoF foot model and the Rizzoli foot model was that we aligned the coordinate system of each rigid body such that when the model is in the anatomical position, the *x*-direction pointed anteriorly, the *y*-direction pointed superiorly and the *z*-direction to the right.

Scaling

Both models (JC and 6-DoF model) were scaled to the dimensions of each participant using markers captured from the static calibration trial, essential for kinematic analysis. In the static trial, participants stood in a comfortable position. The scaling process updates the model's dimensions to customise it to the participant as per the Opensim guidelines (Hicks et al. 2015). Scaling in different dimensions was based on the distance between defined markers on the model compared to those measured in the static trial (see SimTK page for scaling definitions).

Evaluation of the accuracy of the constrained multi-segment model of the foot

Participants

Motion capture and BVR data from seven healthy participants (five males, two females; 21 to 38 yrs) from a previously published study was used for validation purposes. Mean (±S.D.) height and body mass were 166 ± 14 cm and 77 ± 13 kg, respectively. Participants had previously given written consent to partake in the study, which was approved by the University of Queensland and the Providence Medical ethics committees and conducted according to the Declaration of Helsinki.

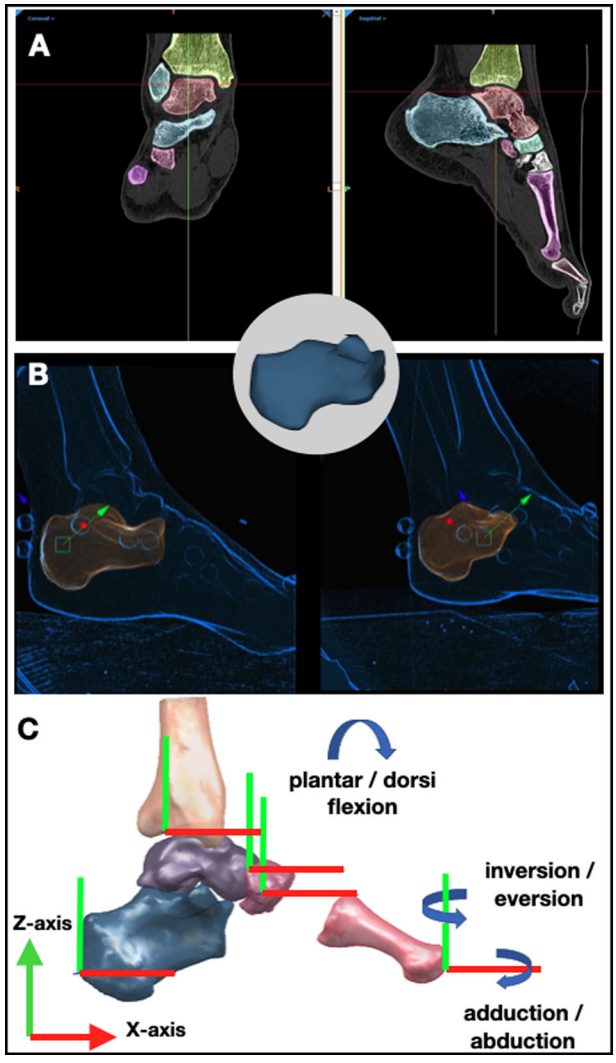


Figure 2. The kinematics of the foot models were compared to subject-specific 3D skeletal motion captured using biplanar videoradiography. This process involved frame by frame registration of 3-dimensional bone models (segmented from CT images, A), to time-series radiographs (B). To compute comparable kinematics between both motion capture systems, the local coordinate system of each bone was aligned to the lab coordinate system at 25% of stance such that the *X*-axis defined the anterior–posterior axis; *Y*-axis defined the medio-lateral axis and the *Z*-axis defined the superior–inferior axis. Motion about the *X*-, *Y*- and *Z*-axes corresponds to adduction and abduction, dorsi- and plantar flexion, inversion and eversion rotation, respectively.

Experimental protocol

Details of the experimental protocol are detailed in Kessler et al. (2019) and Maharaj et al. (2020). Briefly, participants walked and ran at a self-selected pace along a 10 m walkway while high-speed BVR was used to record X-ray images and an opto-electronic motion capture system (Qualysis, Gothenburg, Sweden) was used to record the position of reflective markers of the right foot at 250 Hz simultaneously. One walking and one running trial per participant was used for validation purposes. BVR data were time-synchronized with the opto-electronic motion capture system, which recorded the 3D locations of infrared reflective markers (9 mm) attached to the lower limb and foot (Leardini et al. 2007). Image de-distortion and 3D space calibration of the BVR was performed using previously described methods (Brainerd et al. 2010).

A CT scan of each participant's right foot was captured with the ankle in a plantarflexed orientation. For each scan, the tibia, talus, calcaneus, navicular and first metatarsal were isolated from the entire volume (Figure 2) using threshold and segmentation tools available in Mimics17 (Materialise, Leuven, Belgium).

Data processing

Kinematics from OpenSim

Raw marker positions were recursive filtered at the same frequency using a zero-lag second-order low-pass Butterworth filter with a cut-off frequency of 25 Hz. OpenSim inverse kinematics and body analysis (Delp et al. 2007) was implemented through Matlab (MathWorks, R2018b, Natick, MA, USA) to calculate the relative motion between segments for both the JC and 6-DoF models from trajectories of skin-mounted markers. The process computes the joint angles by stepping through each time frame and positioning the model in a pose that best matches the experimental markers to minimise the difference between the experimental and model marker positions while maintaining JCs (Lu and O'Connor 1999).

Kinematics from biplanar videoradiography

The BVR data were processed using the custom markerless tracking software Autoscoper (Brainerd et al. 2010) (Autoscoper V2, Brown University, Providence, RI, USA). Briefly, the CT volumes for each bone were registered with the BVR data and the bones were manually tracked across all captured frames using the Autoscoper software (Figure 2), as previously described by Kessler et al. (2019) and Maharaj et al. (2020). These techniques track *in vivo* bone motion with an intra-operator reliability of 1.86 mm and 1.90° (Maharaj et al. 2020).

Data analysis and statistics

To compare kinematics acquired from both systems, the global coordinate spaces of the optical motion capture and BVR systems were co-registered using a

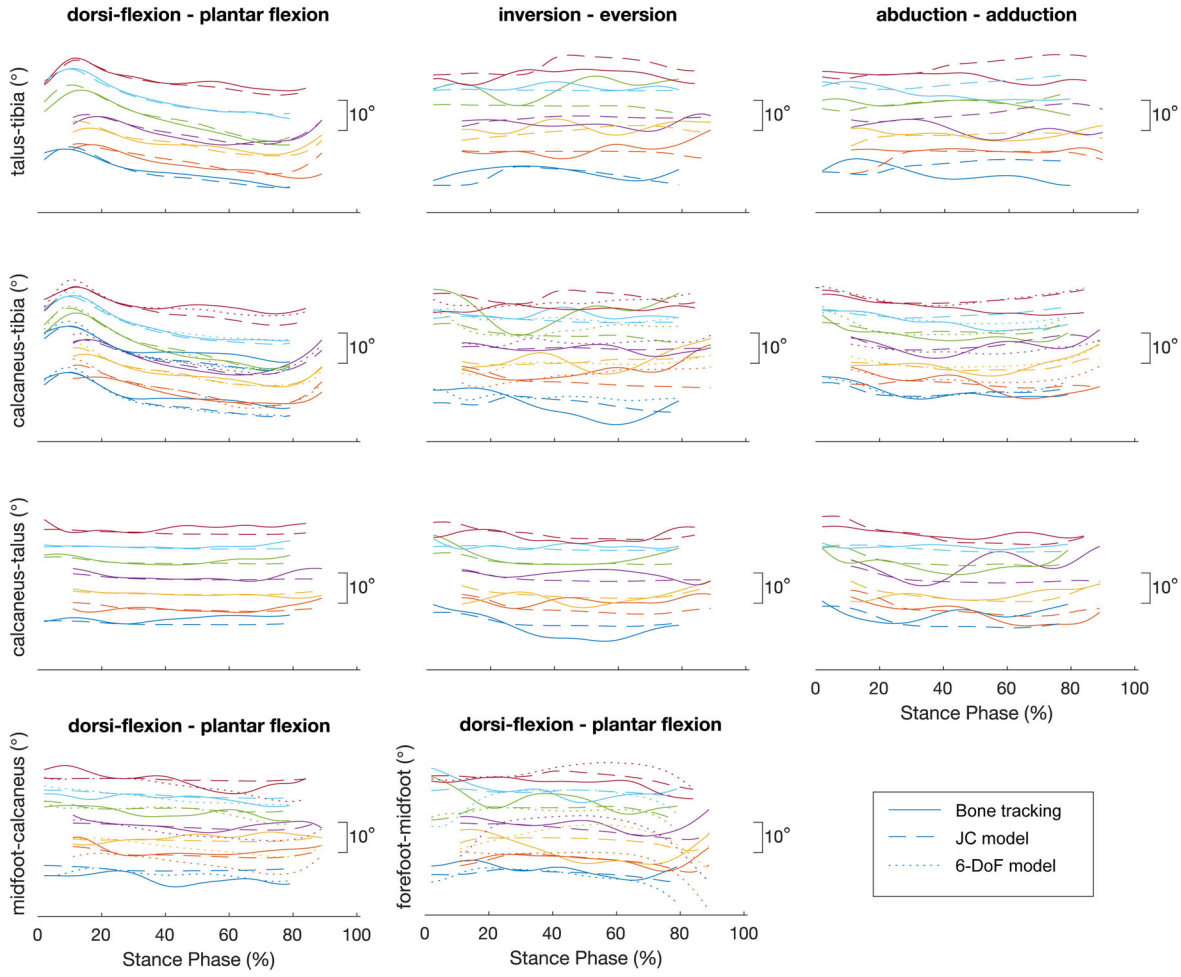


Figure 3. Plots illustrate time normalised kinematics during walking of each participant calculated using biplanar videoradigraphy (solid lines), our joint-constrained (JC) foot model (dashed lines) and a 6-DoF foot model (dotted line). Each colour represents an individual participant. Angles represent displacement relative to the biplanar angle at 25% of the stance phase of gait. Each participant's trace, which represents a single trial, has been vertically offset (by 5°) to reveal the subject-specific similarity between approaches within participants. For the 6-DoF model talus–tibia and calcaneus–talus relative angles were not calcualted as the motion of the talus was fixed rigidly to the calcaneus.

rigid lattice, detailed previously (Miranda et al. 2013). The global coordinate system (laboratory) was subsequently assigned to the geometrical centre of each bone (as defined by the OpenSim model) at 25% of stance phase, when the foot is roughly in a neutral position. As such, the angle of each segment relative to each other at 25% of stance was zero across all dimensions, this allowed comparison of inter-subject joint angles and avoided dependency to an anatomical neutral position of the foot. The relative motions between segments were computed from the transformation matrices describing each segment relative to its more proximal segment and converted to Euler angles using a *YZX* sequence. If we consider that the bone segments are in a roughly neutral position at 25% of stance, foot dorsiflexion/plantar flexion joint rotations occur about the medio-lateral (*Y*) axis in the sagittal plane; inversion/eversion about the vertical (*Z*) axis in the frontal plane; adduction/abduction about the longitudinal (*X*) axis in the transverse plane. Joint rotations of each model and system were filtered using a zero-lag second-order low-pass Butterworth filter with a cut-off frequency of 6 Hz. Each individual's bone pose and joint kinematics relative to time was subsequently time normalised to 101 points across the stance phase, however visual comparisons between models and BVR were only undertaken during the phase when bone motion could be captured in the small BVR capture volume.

For statistical analysis, each trial was cropped to the period of time when there was tracked BVR data for all participants (11–80% of the stance phase). This ensured that the same period of the stance phase was assessed across all participants. Relative motion of the midtarsal (midfoot–calcaneus) and tarsometatarsal

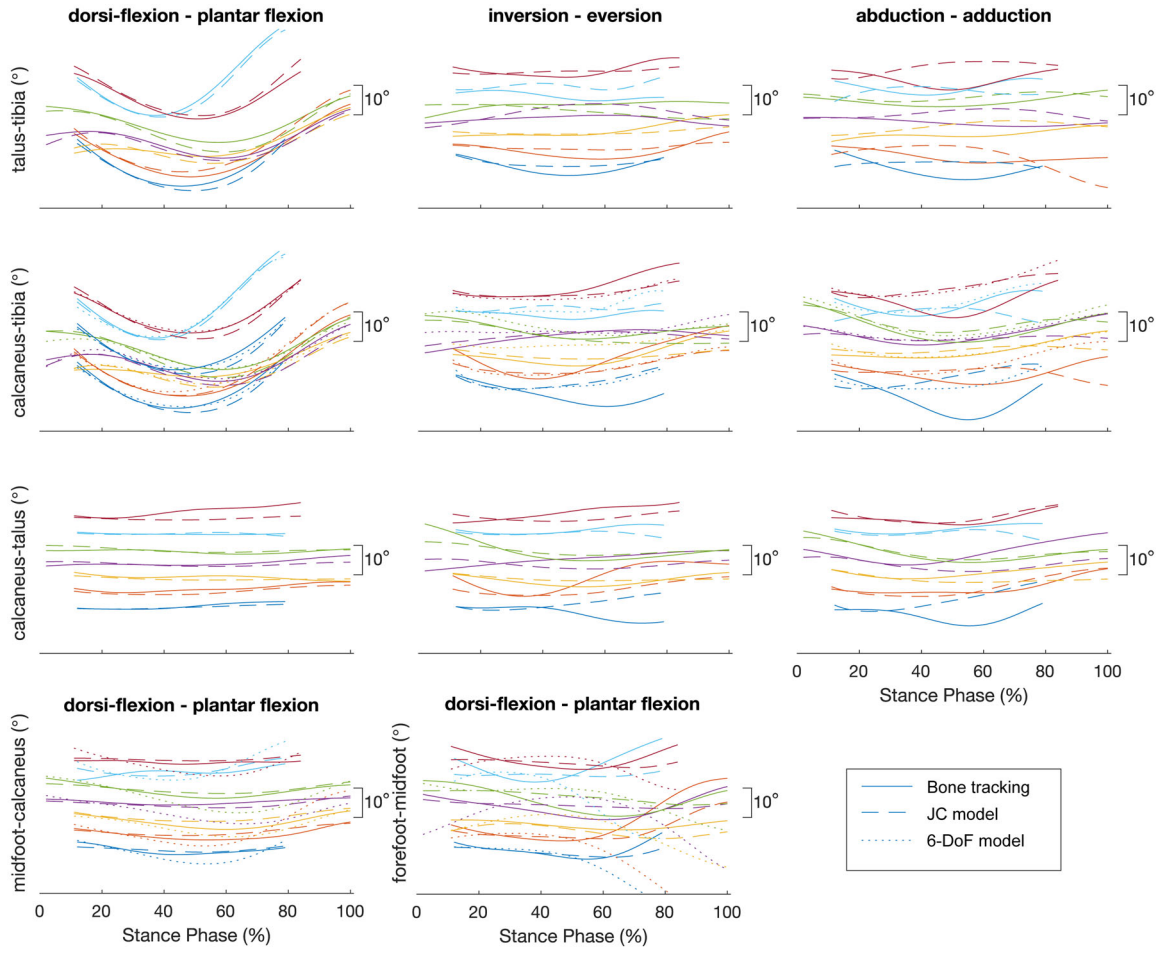


Figure 4. Plots of time-normalised kinematics during running of each participant calculated using biplanar videoradigraphy (solid lines), our joint-constrained (JC) foot model (dashed lines) and a 6-DoF foot model (dotted line). Angles represent displacement relative to the biplanar angle at 25% of the stance phase of gait. Each participant's trace, which represents a single trial, has been offset by 5° to illustrate the similarity between approaches within participants. No talus–tibia and calcaneus–talus relative angles were calcualted for the 6-DoF model as the motion of the talus was fixed rigidly to the calcaneus.

(forefoot–midfoot) joints were assessed in the sagittal plane only due to uncertainties in BVR data. The congruity of models with BVR, at all joints and planes, was compared using the coefficient of multiple correlation (CMC). CMC scores range between 0 and 1 and can be stratified as: poor similarity (0–0.60); moderate (0.60–0.75); good (0.75–0.85); very good (0.85–0.95) and excellent (0.95–1). Root mean square (RMS) differences were calculated to assess the differences in magnitudes across stance. The mean RMS differences across stance for each subject and joint were subsequently compared between models using a paired *t* test.

Sensitivity analysis

A sensitivity analysis was performed to understand the effects of variances in the position of the joint centre of rotation and the orientation of the joint axes on RMS errors. The sensitivity analysis covered a range of variation within the physiologically relevant range of each parameter (Isman and Inman 1969). The *x*, *y* and *z* coordinates of the joint centre of rotation were increased and decreased by 5 and 10 mm while the orientation of the axes were varied by 10° and 20°. The models were systematically modified prior to scaling and inverse kinematic analysis. Intraclass correlation coefficient (ICC) was used to establish reliability and sensitivity of joint parameters to RMS errors.

Results

Performance of multi-segment foot models

Subject-specific kinematics of skeletal motion calculated using BVR (scientific rotoscoping) and for the JC and 6-DoF multi-segment foot models, across all three planes, are illustrated in Figure 3 (walking) and

Table 2. Mean root mean square differences (RMS) and coefficient of multiple correlation (CMC) and their ranges for our joint constrained (JC) model and the 6-DoF model, as compared to biplanar videoradigraphy across the stance phase for each gait.

	RMS errors (mm)		CMC	
	JC	6-DoF	JC	6-DoF
		WALKING		
		dorsi - plantar flexion		
talus-tibia	0.99 (0.6–1.14)		0.98 (0.97–1)	
calcaneus-tibia	1.5 (0.8–2.81)	1.42 (0.5–2.76)	0.94 (0.79–0.99)	0.95 (0.86–1)
calcaneus-talus	1.18 (0.43–2.04)		0.28 (0–0.77)	
midfoot-calcaneus	1.93 (1.24–3.47)	2.8 (1.36–4.54)	0.31 (0–0.7)	0.43 (0–0.75)
forefoot-midfoot	2.48 (0.8–5.45)	4.38 (1.93–8.62)	0.37 (0–0.93)	0.16 (0–0.69)
		inversion - eversion		
talus-tibia	2.95 (1.39–6.96)		0.11 (0–0.62)	
calcaneus-tibia	3.73 (1.1–8.33)	3.14 (1.77–6.05)	0.13 (0–0.56)	0.18 (0–0.73)
calcaneus-talus	2.13 (1.11–3.02)		0.24 (0–0.75)	
		abduction - adduction		
talus-tibia	4.14 (2.01–6.05)		0 (0–0)	
calcaneus-tibia	2.44 (1.25–3.9)	2.35 (1.58–3.79)	0.37 (0–0.75)	0.42 (0–0.76)
calcaneus-talus	2.74 (1.31–5.33)		0.27 (0–0.83)	
		RUNNING		
		dorsi - plantar flexion		
talus-tibia	1.88 (1.09–2.67)		0.93 (0.73–1)	
calcaneus-tibia	1.91 (1.22–2.96)	1.32 (0.52–2.45)	0.92 (0.63–1)	0.96 (0.82–1)
calcaneus-talus	0.99 (0.43–2.72)		0.34 (0–0.87)	
midfoot-calcaneus	1.27 (0.81–1.9)	2.89 (1.74–4.69)	0.46 (0–0.83)	0.53 (0–0.82)
forefoot-midfoot	2.98 (1.83–5.17)	7.41 (2.82–15.9)	0.41 (0–0.71)	0.07 (0–0.37)
		inversion - eversion		
talus-tibia	2.27 (1.01–3.74)		0.25 (0–0.73)	
calcaneus-tibia	2.72 (1.02–6.14)	3.46 (1.96–5.48)	0.53 (0–0.89)	0.32 (0–0.84)
calcaneus-talus	2.87 (1.3–5.09)		0.27 (0–0.84)	
		abduction - adduction		
talus-tibia	3.74 (2.2–5.87)		0 (0–0)	
calcaneus-tibia	4.38 (1.25–9.96)	3.3 (1–7.12)	0.29 (0–0.76)	0.52 (0–0.95)
calcaneus-talus	2.65 (0.67–6.53)		0.37 (0–0.98)	

The JC model performed significantly ($p < 0.01$) better, with smaller mean RMS errors across both gaits, planes and comparable joints compared to the 6-DoF model [JC: 2.19° (1.43 to 2.73); 6-DoF: 3.25° (1.37 to 5.89)]. In particulary, the JC model demonstrated significantly ($p < 0.01$) smaller errors in the sagittal plane compared to the 6-DoF model [JC: 1.92° (1.43 to 2.73); 6-DoF: 3.37° (1.37 to 5.89)] across comparable joints. Analysis was only done from 11% to 80% of stance.

Figure 4 (running). Overall, the JC model exhibited significantly ($p < 0.01$) smaller mean RMS differences compared to the 6-DoF model [JC: 2.19° (1.43 to 2.73); 6-DoF: 3.25° (1.37 to 5.89)] while CMC scores ranged from poor to excellent for both models (JC: 0.25 to 0.96; 6-DoF: 0.12 to 0.96) across comparable joints and planes. The JC model had significantly smaller RMS differences in the sagittal plane [JC (talus-tibia, midfoot-calcaneus, forefoot-midfoot): 1.92° (1.43 to 2.73); 6-DoF (calcaneus-tibia, midfoot-calcaneus, forefoot-midfoot): 3.37° (1.37 to 5.89)]. The largest mean differences across all gaits and planes in the JC model were evident between the forefoot-midfoot [2.73° (0.8 to 5.45) followed by the talus-tibia [2.66° (1.43 to 3.94)], calcaneus-talus [2.09° (1.09 to 2.7)] and midfoot-calcaneus [1.6° (0.81 to 3.47)]. The largest mean differences across all gaits and planes in the 6-DoF model were also evident between the forefoot-midfoot [5.89° (1.93 to 15.9)]. Differences between the midfoot-calcaneus and calcaneus-tibia of the 6-DoF model were 2.85° (1.36 to 4.69) and 2.5° (1.37 to 3.3), respectively.

Ankle joint

Ankle kinematics were calculated as the motion of the talus relative to the tibia for the BVR data and JC model as this represents the true morphological joint. In the 6-DoF model however, the talus and calcaneus were rigidly fixed (by necessity, as there are no markers on the talus), therefore, ankle kinematics are represented by the motion between the calcaneus and tibia. There were no differences in RMS errors between the models in the sagittal [JC: 1.43° (0.6 to 2.67); 6-DoF: 1.37° (0.5 to 2.76)], frontal [JC: 2.61° (1.01 to 6.96); 6-DoF: 3.3° (1.77 to 6.05)] and transverse planes [JC: 3.94° (2.01 to 6.05); 6-DoF 2.82° (1.0 to 7.12)]. The similarity of the curves for both models compared to BVR ranged from very good to excellent for sagittal plane motion [JC: 0.73 to 1); 6-DoF 0.82 to 1)] but ranged from poor to very good for frontal and transverse plane motions, see Table 2.

Subtalar joint

The mean RMS differences in the relative motion between the calcaneus and talus between the JC

model and BVR tracking, across all planes, was 2.09°; the errors in the sagittal, frontal and transverse planes were: 1.09° (0.43 to 2.72); 2.5° (1.11 to 5.09); 2.7° (0.67 to 6.53), respectively. The CMC scores suggested that the similarity between the curves ranged from poor to excellent across all different planes. The ranges of the CMC scores for subtalar joint sagittal, frontal and transverse planes motions were: 0 to 0.87; 0 to 0.84; 0 to 0.98, respectively. Subtalar joint kinematics of the 6-DoF model could not be calculated as the motion of the talus was fixed rigidly to the calcaneus.

Midtarsal joint

Segmental motion of the midfoot calculated from the JC and 6-DoF models was compared to the navicular skeletal motion tracked from BVR. Significant ($p < 0.01$) mean RMS differences were found for midfoot sagittal plane motion between the JC and 6-DoF models, such that the JC model demonstrated smaller differences compared to the 6-DoF model [JC: 1.6° (0.81 to 3.47); 6-DoF: 2.85° (1.36 to 4.69)]. The similarity in the waveforms of the JC model and 6-DoF compared to BVR ranged from 0 to 0.83 and 0 to 0.82, respectively.

Tarsometatarsal joint

Similar to the midfoot, forefoot motion of the JC and 6-DoF models was compared to kinematics of the first metatarsal bone tracked from BVR. The JC model demonstrated significantly ($p < 0.01$) smaller mean RMS differences in the sagittal plane motion compared to the 6-DoF model [JC: 2.73° (0.8 to 5.45); 6-DoF: 5.89° (1.93 to 15.9)]. The CMC scores indicated that the similarity in waveforms was also greater for the JC model ($p < 0.01$), ranging from weak to very good (0 to 0.93), compared to the weak to moderate similarity (0 to 0.69) between the 6-DoF and BVR data.

Sensitivity analysis

The ICC values comparing variations in the position of the joint centre of rotation based on RMS errors ranged between 0.985 and 0.998 for all joints (see Supplementary Data). RMS differences were most sensitive to the position of the centre of rotation for ankle joint transverse plane (abduction and adduction) motion, with an ICC of 0.985. Subtalar joint RMS differences were most sensitive to variations in the joint axes with an ICC of 0.959–0.976. The ICC values for the orientation of the joint axes on the midtarsal and tarsometatarsal joints ranged from 0.965 to 0.990.

Discussion

We created an articulated multi-segment foot and ankle model that is capable of performing musculoskeletal analyses of human walking and running. The model reported kinematics of the human foot during the stance phase of locomotion with a mean RMS difference of 2.2° across all gaits, planes and comparable joints compared to BVR, and outperformed the 6-DoF model (mean RMS difference = 3.3°). The mean errors of the JC model were also less than those comparing a 6-DoF model to bone pin data (3.3° for the same comparable joints) (Nester et al. 2007). In addition to improved accuracy, the JC foot model is capable of quantifying subtalar joint motion and yields intrinsic foot kinematics that are physiologically more realistic; tri-planar motion of combined dorsiflexion–inversion–adduction and plantarflexion–eversion–abduction rather than motion in six independent degrees of freedom. Therefore, the intrinsic foot kinematics computed from the model may be easier to interpret and compare across participants. In this study we report bone to bone kinematics to compare the models to BVR for accuracy. Output joint angles of the subtalar joint have previously been published, see Maharaj et al. (2016).

The improved ability of the JC model to estimate kinematics within the foot, as compared to the 6-DoF model, may be due to different factors. First, the reflective markers are subject to soft-tissue motion-artefacts and associated measurement-error (Kessler et al. 2019). Due to these errors, motion described by the 6-DoF model may be much larger than the motion that would be realistically possible. In contrast, the JC model consists of segments interconnected by joints, which define how each segment moves with respect to its more proximal neighbour. JCs minimise effects of the soft-tissue artefacts on motion-capture markers and associated measurement-error, and result in more realistic motion (Lu and O'Connor 1999). Second, the constraints may better represent the structural stability of intrinsic joints in the foot that are imposed by bone morphology (Hicks 1953; Bruckner 1987). While, constraining joints may preclude an understanding of how articular, passive and active tissues interact to create joint stability, the limited degrees of freedom in the JC model is better suited for musculoskeletal simulations, whereby an

equilibrium solution may be easier to find. The JC model may also be used for analyses of soft-tissue dynamics, specifically muscle-tendon length changes, because such analyses are primarily driven by joint angular kinematics.

The variances observed in the JC model kinematics and BVR data may arise from multiple sources, including the requirement to use skin-mounted markers to track motion, limitations in the rigid body assumptions in the JC model, and potential inaccuracies in BVR bone pose estimation using scientific rotoscoping. Unfortunately, whilst we suspect the list above is in order of importance, the current study design does not allow us to distinguish which factor has the greatest effect on the RMS differences. The differences are likely to be, at least partially, due to soft-tissue artefact. Using a similar data set, Kessler et al. (2019) compared virtual markers driven by BVR skeletal motion to skin-mounted reflective markers and found the greatest differences in the medial–lateral direction (mean RMS errors of 2.9 mm during walking and running). The differences evident in the distal joints are likely to be due to violation of the rigid body assumption and the definition of rigid functional segments; motion of these segments were driven by markers on multiple bones but compared to skeletal motion of an individual bone.

The differences between the BVR and JC model outputs are also likely to be associated with inadequacies of BVR, primarily the restricted field of view and element occlusion, which limits accurate capture of the motion of the foot bones, particularly in the frontal plane. The accuracy of tracking skeletal function during *in vivo* gait with scientific rotoscoping remains unknown. A reliability paper, performed on the same dataset used here, illustrated that manually tracked BVR data is most reliable in the sagittal plane, followed by frontal and transverse planes. The mean intra-operators RMS errors in bone orientations for sagittal, frontal and transverse planes were 1.42°, 2.12° and 2.14°, respectively (Maharaj et al. 2020). Our data also show inconsistent pattern of kinematics in the BVR data in the frontal and transverse planes (Figures 3 and 4), and in the distal segments. This is likely due to the manual process of extracting skeletal motion *via* scientific rotoscoping or small errors in calibration. Furthermore, some of the 'wiggle' in the BVR data through the gait cycle may be a by-product of the spline function that was used to interpolate between manually tracked keyframes, which may have contributed to the low CMC scores. Thus, we have been careful to not claim the RMS differences as errors as we currently do not understand the accuracy of the BVR technique used here.

Errors in foot kinematics may also arise from the scaling process for each model. The initial scaling process influences the location of the centre of rotation of the joint and has been shown to have significant effect on the accuracy of lower limb kinematics, kinetics and joint contact forces. While our sensitivity analysis demonstrates that the centre of rotation has a minimal effect in the intrinsic joints in the foot except for the subtalar joint, better personalisation of the model through more sophisticated scaling methods such as statistical shape modelling (Grant et al. 2020) may further improve the performance of the musculoskeletal model. Finally, differences may arise from the definition and number of the rigid segments in the model. It is possible that division of the foot along the frontal plane (calcaneus, midfoot, forefoot and digits) may not sufficiently represent movements that occur along the frontal plane (inversion/eversion). Thus, perhaps, more segments/joints are required to better capture frontal and transverse plane motion, particularly segments that are created by dividing the foot in the sagittal plane, that is, medial and lateral segments (MacWilliams et al. 2003). It is also possible that errors in one segment are transmitted to more distal segments, illustrating a limitation of this approach when constraints are enforced.

The JC model was created upon the assumption that no translation occurs between neighbouring/adjacent segments. The calcaneus does, however, translate relative to the talus. For example, a recent study reported a translation of 0.7 mm ± 0.46 mm in the centre of rotation of the calcaneus during inversion and eversion motion using weight-bearing clinical CT (Fernández et al. 2020). As such, we examined the sensitivity of the model outputs to movement of the joint centre and axis. We found that subtalar joint kinematics were sensitive to variation in the location of the centre of rotation. However, the overall effect on RMS differences compared to BVR was small (±1°). A limitation of this study is that we were unable to assess the effects of model choice and constraints on kinetics (muscle, ligament, contact, etc.) in a straightforward way. Incorporating segment translations, never-the-less, remains the ideal case scenario (Fernández et al. 2020). However, it requires the development of technologies such as BVR to model the relationship between rotations and translations of segments accurately to subsequently incorporate into models, in combination with increased sensitivity of motion capture technologies.

Conclusion

We have developed a multi-segment foot model, in which motion between segments are constrained to emulate the role of intrinsic joints in the foot, and demonstrated that this model was capable of accurate musculoskeletal analyses of human walking and running. Our JC foot model was able to compute joint angles with greater similarity (particularly in the sagittal plane) to measurements obtained *via* BVR, when compared to a typically implemented 6-DoF model in relatively typical adult feet. In addition, the JC model was capable of quantifying motion between the calcaneus and talus (subtalar joint), yielding intrinsic foot kinematics with physiologically realistic triplane motion, and may also be used for analyses of soft-tissue dynamics.

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Disclosure statement

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